



Mark Scheme (Results)

Summer 2026

Pearson Edexcel Advanced Extension Award
In Mathematics (9811)
Paper 01

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General Marking Guidance

These notes offer general guidance, but the specific notes for examiners appertaining to individual questions take precedence.

- 1 All candidates must receive the same treatment. Examiners must mark the last candidate in exactly the same way as they mark the first.

Where some judgement is required, mark schemes will provide the principles by which marks will be awarded; exemplification/indicative content will not be exhaustive. When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the response should be sent to review.

- 2 All the marks on the mark scheme are designed to be awarded; mark schemes should be applied positively. Examiners should also be prepared to award zero marks if the candidate's response is not worthy of credit according to the mark scheme. If there is a wrong answer (or no answer) indicated on the answer line always check the working in the body of the script (and on any diagrams), and award any marks appropriate from the mark scheme.

Questions where working is not required: In general, the correct answer should be given full marks.

Questions that specifically require working: In general, candidates who do not show working on this type of question will get no marks – full details will be given in the mark scheme for each individual question.

- 3 **Crossed out work**

This should be marked **unless** the candidate has replaced it with an alternative response.

- 4 **Choice of method**

If there is a choice of methods shown, mark the method that leads to the answer given on the answer line.

If no answer appears on the answer line then mark both methods **as far as they are identical** and award these marks.

- 5 **Incorrect method**

If it is clear from the working that the “correct” answer has been obtained from incorrect working, award 0 marks.

6 Follow through marks

Follow through marks which involve a single stage calculation can be awarded without working as you can check the answer, but if ambiguous do not award.

Follow through marks which involve more than one stage of calculation can only be awarded on sight of the relevant working, even if it appears obvious that there is only one way you could get the answer given.

7 Ignoring subsequent work

It is appropriate to ignore subsequent work when the additional work does not change the answer in a way that is inappropriate for the question or its context. (eg an incorrectly cancelled fraction when the unsimplified fraction would gain full marks).

It is not appropriate to ignore subsequent work when the additional work essentially makes the answer incorrect (eg incorrect algebraic simplification).

8 Probability

Probability answers must be given as a fraction, percentage or decimal. If a candidate gives a decimal equivalent to a probability, this should be written to at least 2 decimal places (unless tenths). Incorrect notation should lose the accuracy marks, but be awarded any implied method marks.

If a probability fraction is given then cancelled incorrectly, ignore the incorrectly cancelled answer.

9 Range of answers

Unless otherwise stated, when an answer is given as a range (eg 3.5 – 4.2) then this is inclusive of the end points (eg 3.5, 4.2) and all numbers within the range.

Guidance on the use of abbreviations within this mark scheme

M	method mark awarded for a correct method or partial method
A	accuracy mark (awarded after a correct method; if no method is seen then full marks for the question are implied but see individual mark schemes for more details)
B	unconditional accuracy mark (no method needed)
oe	or equivalent
cao	correct answer only
ft	follow through (when appropriate as per mark scheme)
sc	special case
dep	dependent (on a previous mark)
indep	independent
awrt	answer which rounds to
isw	ignore subsequent working

Question Number	Scheme		Marks
1	$\left(\frac{dy}{dx} = \lim_{h \rightarrow 0}\right) \frac{\frac{1}{\sin(x+h)} - \frac{1}{\sin x}}{h}$	Writes down/uses the correct gradient of the chord. May be in terms of cosec x .	B1
	$\frac{\frac{1}{\sin(x+h)} - \frac{1}{\sin x}}{\sin(x+h) - \sin x} = \frac{\sin x - \sin(x+h)}{\sin x \sin(x+h)}$ $= \frac{\sin x \pm \sin x \cos h \pm \cos x \sin h}{\sin x \sin(x+h)}$	Puts over a common denominator and attempts to use the addition formula in the numerator (with or without the h)	M1
	$\left(\frac{dy}{dx} = \lim_{h \rightarrow 0}\right) \frac{\sin x - (\sin x \cos h + \cos x \sin h)}{h \sin x \sin(x+h)}$	Correct expression with a common denominator, the numerator expanded and h included in the denominator. Accept with nested fractions for this mark. Denominator may also be expanded.	A1
	$\left(= \lim_{h \rightarrow 0}\right) \frac{1 - \cos h}{h} \times \frac{\sin x}{\sin x \sin(x+h)}$ $- \frac{\sin h}{h} \times \frac{\cos x}{\sin x \sin(x+h)}$	Factorises and extracts/identifies the standard limits within their expression – must make the individual limits clear for this mark (should be separate fractions).	dM1
	As $h \rightarrow 0$, $\frac{\sin h}{h} \rightarrow 1$, $\frac{\cos h - 1}{h} \rightarrow 0$ (and $\sin(x+h) \rightarrow \sin x$) so $\frac{dy}{dx} = 0 \times \frac{1}{\sin x} - 1 \times \frac{\cos x}{\sin x \sin x}$	Use of both given limits applied at an appropriate time. Note this mark may be scored for correct application implied even if the limits are not made explicitly clear.	B1
	$\frac{dy}{dx} = -\operatorname{cosec} x \cot x^*$	Fully correct proof with correct notation throughout and use of limits clearly evident.	A1*
			(6)
(Total 6 marks)			

Question Number	Scheme		Marks
2(a)	$f(x) = \left(x - \frac{3}{2}\right)^2 + \dots$ or $f'(x) = 2x - 3 = 0 \Rightarrow x = \dots$	Attempts to complete the square (may be implied). Alternatively, differentiates, sets = 0 and attempts to solve.	M1
	$(k =) \frac{3}{2}$ condone $k \leq \frac{3}{2}$ or $x \leq \frac{3}{2}$	Deduces the correct value of k . The k need not be given but $x = \frac{3}{2}$ is A0. Answer only scores both marks (without clearly incorrect work).	A1
			(2)
(b)	$y = \left(x - \frac{3}{2}\right)^2 + " \frac{11}{4} " \rightarrow x = \dots$ or $x = \left(y - \frac{3}{2}\right)^2 + " \frac{11}{4} " \rightarrow y = \dots$	Sets $y = \dots$ and attempts to make x the subject of their completed the square form condoning sign errors (or vice versa)	M1
	$f^{-1}(x) = \frac{3}{2} - \sqrt{x - \frac{11}{4}}$ or e.g. $f^{-1}(x) = \frac{3 - \sqrt{4x - 11}}{2}$	Correct inverse function in any form but LHS must be correct notation with $f^{-1}(x)$, not y . Must have selected the correct root.	A1
	$x \geq " \frac{11}{4} "$	Correct domain following through on their $\frac{11}{4}$. Must be $x \geq$	B1ft
			(3)
(c)	$h^{-1}(x) = \sqrt{\frac{3 - \sqrt{4x - 11}}{2}}$ or e.g. $h^{-1}(x) = \sqrt{\frac{3}{2} - \sqrt{x - \frac{11}{4}}}$	Deduces a correct form for h^{-1} following through on their f^{-1} . Condone $y = \dots$ here. ($\pm$ included is B0)	B1ft
	$" \frac{11}{4} " \leq x \leq 5$	Correct domain following through on their $\frac{11}{4}$. (Must be using x , not h^{-1}) Attempts to restart via completing the square or differentiation are acceptable for both marks	B1ft
			(2)
(Total 7 marks)			

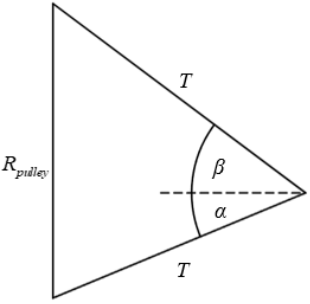
Question Number	Scheme		Marks
3(a)	$P(X = 7) = \frac{1}{9} \cdot \frac{1}{243} = \frac{1}{2187} \quad \text{and}$ $P(X = 8) = \frac{1}{5\sqrt{5}} a = \frac{\sqrt{5}}{25} a \quad (o.e.)$	Both correct, accept unsimplified and isw	B1
			(1)
(b)	$S_{\infty}(\text{odd}) = \frac{\frac{1}{3}}{1 - \frac{1}{9}} \left(= \frac{3}{8} \right) \text{ or } S_{\infty}(\text{even}) = \frac{a}{1 - \frac{1}{\sqrt{5}}}$	Attempts the sum to infinity for either the odd or even k	M1
	$S_{\infty}(\text{even}) = \frac{a}{1 - \frac{1}{\sqrt{5}}} = 1 - \frac{3}{8} \Rightarrow a = \frac{5 - \sqrt{5}}{8} *$	Uses that the probabilities sum to 1 to sets up and solve an equation in a leading to the given answer.	A1*
(c)			(2)
	$P(1 < X \leq 8 \cap X \text{ is even}) =$ $a + a \times \frac{1}{\sqrt{5}} + a \times \left(\frac{1}{\sqrt{5}} \right)^2 + a \times \left(\frac{1}{\sqrt{5}} \right)^3 = \dots \left\{ \frac{3}{5} \right\}$ or $\frac{5 - \sqrt{5}}{8} \left(1 - \left(\frac{1}{\sqrt{5}} \right)^4 \right) / \left(1 - \frac{1}{\sqrt{5}} \right) = \dots \left\{ \frac{3}{5} \right\}$	Attempts to find $P(1 < X \leq 8 \cap X \text{ is even})$ using the given a or in terms of a via a correct method.	M1
	$P(1 < X \leq 8 X \text{ is even})$ $= \frac{P(1 < X \leq 8 \cap X \text{ is even})}{P(X \text{ is even})}$ $= \frac{a + a \times \frac{1}{\sqrt{5}} + a \times \left(\frac{1}{\sqrt{5}} \right)^2 + a \times \left(\frac{1}{\sqrt{5}} \right)^3}{1 - \frac{3}{8}} = \dots$ or $= \frac{\frac{5 - \sqrt{5}}{8} \left(1 - \left(\frac{1}{\sqrt{5}} \right)^4 \right)}{1 - \frac{1}{\sqrt{5}}} \times \frac{1}{1 - \frac{3}{8}} = \dots$	Correct method to find the required probability using the given a . Must achieve a rational answer. May use the sum to infinity to find $P(X \text{ even})$ in which case the a 's will cancel, so may not see the substitution.	dM1
	$= \frac{24}{25}$	cao	A1
			(3)
(d)	$P(Y < 5) = P(Y = 3) + P(Y = 4)$	Identifies the required probabilities (may be implied)	B1
	$= \left(\frac{1}{3} \right)^3 + \frac{5 - \sqrt{5}}{8} \times \left(\frac{1}{3} \right)^2 \times 3$	Correct method, must have the three cases for $Y = 4$. May be in terms of a .	M1
	$= \frac{53 - 9\sqrt{5}}{216}$	cao in simplest form	A1
			(3)
(Total 9 marks)			

Question Number	Scheme		Marks
	Note: If they use the result of part (iii) (and (ii)) to do part (i) then send to review.		
4(a)(i)	$\frac{y-k}{A} + y^2 = r^2$ or $Ay^2 + y - k = Ar^2$	Substitutes to eliminate x	M1 (S+)
	$Ay^2 + y - k - Ar^2 = 0$ Alt: $\left(y + \frac{1}{2A}\right)^2 - \frac{1}{4A^2} - \frac{k}{A} = r^2$	Correct 3TQ in y set = 0. Alternatively, allow for a correct complete square equation.	A1
	$1 - 4A(-k - Ar^2) = 0$ or $r^2 + \frac{k}{A} + \frac{1}{4A^2} = 0$	Attempts the discriminant = 0 from either their 3TQ with gathered terms or via completing the square.	dM1
	$k = -\frac{1}{4A} - Ar^2 *$	Cancels terms and reaches the given answer with no errors. Equality must be used throughout.	A1*
			(4)
Alt (a)(i)	$x^2 + (Ax^2 + k)^2 = r^2$ $x^2 + A^2x^4 + 2Akx^2 + k^2 = r^2$	Substitutes to eliminate y and attempts to expand.	M1
	$A^2x^4 + (2Ak + 1)x^2 + k^2 - r^2 = 0$ or $A^2\left(x^2 + \frac{1 + 2Ak}{2A^2}\right)^2 - \frac{(1 + 2Ak)^2}{4A^2} + k^2 = r^2$	Correct 3TQ in x^2 set = 0 or correctly completes square on x^4	A1
	$(2Ak + 1)^2 - 4A^2(k^2 - r^2) = 0$ or $\frac{(1 + 2Ak)^2}{4A^2} - k^2 + r^2 = 0$	Attempts the discriminant = 0 from either their 3TQ with gathered terms or via completing the square.	dM1
	$4A^2k^2 + 4Ak + 1 - 4A^2k^2 + 4A^2r^2 = 0$ $k = -\frac{1}{4A} - Ar^2 *$	Cancels terms and reaches the given answer with no errors. Equality must be used throughout.	A1*
			(4)
Alt II	$\frac{dy}{dx} = 2Ax \Rightarrow m_N = \frac{-1}{2Ax_1}$ $\Rightarrow y - y_1 = \frac{-1}{2Ax_1}(x - x_1)$ Passes through (0, 0) $\Rightarrow y_1 = -\frac{1}{2A}$ $\Rightarrow x_1^2 = -\frac{1}{2A^2} - \frac{k}{A}$ Alt: $2x + 2y\frac{dy}{dx} = 0 \Rightarrow 2Ax_1 = -\frac{x_1}{y_1}$ etc	Sets up the normal at a general point via correct work and uses that it passes through (0,0) to find y and x^2 . Alt: may compare gradients (S+).	M1
		Correct coordinates (may imply M1A1 for (ii))	A1
	$\dots \Rightarrow r^2 = -\frac{1}{2A^2} - \frac{k}{A} + \frac{1}{4A^2}$ $k = -\frac{1}{4A} - Ar^2 *$	Substitutes into circle equation and shows results per other methods.	dM1 A1*
			(4)
Alt III	$d^2 = x^2 + y^2 = x^2 + (Ax^2 + k)^2$ $\frac{dd^2}{dx} = 0 \Rightarrow 2x + 2(Ax^2 + k) \cdot 2Ax = 0$ $(x \neq 0 \Rightarrow)x^2 = \dots$	Attempts a process to minimise distance from point on C_1 to O to find expression for x^2	M1
	$x^2 = -\frac{1}{2A^2} - \frac{k}{A}$	Correct expression for x^2 in terms of k and A	A1

Question Number	Scheme		Marks
	$r^2 = x^2 + y^2 = -\frac{1}{2A^2} - \frac{k}{A} + \left(-\frac{1}{2A} - k + k\right)^2$	Substitutes into circle equation to get equation relating k , A and r	dM1
	$r^2 = -\frac{1}{2A^2} - \frac{k}{A} + \frac{1}{4A^2}$ $k = -\frac{1}{4A} - Ar^2 *$	Cancels terms and reaches the given answer with no errors. Equality must be used throughout.	A1*
			(4)
(a)(ii)	$Ay^2 + y - \left(-\frac{1}{4A} - Ar^2\right) - Ar^2 = 0$ $y^2 + \frac{1}{A}y + \frac{1}{4A^2} = 0 \Rightarrow y = \dots$	Full method to find y using k . E.g. substitutes the given answer into a 3TQ in terms of y and simplifies to eliminate r , or any other complete method (e.g. substitutes x^2 into equation for C_1 via Alt II). Note use of part (iii) without use of (i) is M0A0.	M1
	$y = -\frac{1}{2A}$	Cao A0 if positive case also included.	A1
			(2)
(a)(iii)	$x^2 + \left(\frac{-1}{2A}\right)^2 = r^2$ $x = \pm\sqrt{r^2 - \frac{1}{4A^2}} *$	Substitutes $y = -\frac{1}{2A}$ into $x^2 + y^2 = r^2$ and makes x the subject or other full method to establish the result. Must have come from correct (negative) y coordinate, but allow if $\pm$ had been used in (ii).	B1*
			(1)
(a)(iv)	$r > \frac{1}{2A}$	cao	B1
			(1)
(b)(i)	e.g. $y \pm b = A(x \pm a)^2 + k$	Attempts to replace x with $x \pm a$ and y with $y \pm b$ with or without k replaced using (a)(i)	M1
	$y = A(x - a)^2 - \frac{1}{4A} - Ar^2 + b$ $(y = Ax^2 - 2aAx + a^2A - \frac{1}{4A} - Ar^2 + b)$	Correct equation in the form $y = \dots$ with k replaced using (a)(i). No need to expand and isw after a correct equation.	A1
			(2)
(b)(ii)	$x = a$	Deduces $x = a$ is normal to both curves	B1
	Gradient $= \frac{-\frac{1}{2A} + b - b}{\pm\sqrt{r^2 - \frac{1}{4A^2}} + a - a} \left(= \mp \frac{1}{\sqrt{4A^2r^2 - 1}} \right)$ Or e.g. $C_1: \frac{dy}{dx} = 2Ax \Rightarrow m_N = \frac{-1}{\pm 2A\sqrt{r^2 - \frac{1}{4A^2}}}$	Attempts the gradient using their y and the given x for the original situation or with centre (a, b) and ignore signs. May also come from differentiating either the circle or the quadratic – before or after transformation – but must be using negative reciprocal gradient (but give bod if unclear).	M1 (S+)

Question Number	Scheme		Marks
	$y-b=\frac{1}{\sqrt{4A^2r^2-1}}(x-a)$	One of the other normal equations in any form. Accept with $\pm$'s as long as a correct equation is clear.	A1
	or/and $y-b=-\frac{1}{\sqrt{4A^2r^2-1}}(x-a)$	Both of the normal equations in any form and no others (other than $x=a$). Any uses of multiple $\pm$ and $\mp$ must be consistent.	A1
			(4)
	Award S1 for: • a fully correct solution that is succinct but does not mention any S+ points • solution scoring 11+ marks that is succinct and includes at least one S+ point • solution scoring 11+ marks that may be laboured but includes both S+ points		S1
			(Total 14+1 marks)
Notes: S+ for eliminating x rather than y in (a)(i) / comparing gradients S+ for using the gradient of the radius with (a)(ii) and (a)(iii) in (b) i.e. no differentiation or realising the transformation does not affect the gradient so uses the original equation.			

Question Number	Scheme	Marks
5(a)	Note: Use of $F = ma$ (in the parallel directions) can score the second and third Ms but will require $a = 0$ to be used to recover further marks. Condone missing g's on some (or all) terms for the M's.	
	For block A: $\begin{cases} \nearrow R_A = 26mg \cos \alpha (= 24mg) \\ \nearrow T = 26mg \sin \alpha + F_A \end{cases}$ $\left(T = 10mg + F_A = 10mg + \frac{1}{2}(24mg) = 22mg \right)$ OR $R_A \cos \alpha + (T - F_A) \sin \alpha = 26mg$ or $5R_A + 12(T - F_A) = 26mg$ $R_A \sin \alpha - (T - F_A) \cos \alpha = 0$ or $12R_A - 5(T - F_A) = 0$	M1
	Correct equations but allow any inequality for the = if using μR May evaluate the trig terms.	A1
	For block B: $\begin{cases} \nearrow R_B = 75mg \cos \beta + P \sin \beta \\ \nwarrow T + P \cos \beta + F_B = 75mg \sin \beta \\ \text{or } \nwarrow (T) + P \cos \beta - F_B = 75mg \sin \beta \end{cases}$ (as friction could be acting in either direction depending on the size of P) $\left(\begin{aligned} &\nearrow R_B = 60mg + \frac{3}{5}P \\ &\nwarrow "22mg" + \frac{4}{5}P + \frac{1}{5}R_B = 45mg \\ &\text{or } \nwarrow ("22mg") + \frac{4}{5}P - \frac{1}{5}R_B = 45mg \end{aligned} \right)$ OR $R_B \cos \beta + (T - F_B) \sin \beta = 75mg$ or $4R_B + 3(T - F_A) = 75mg$ $R_B \sin \beta - (T - F_B) \cos \beta = P$ or $3R_B - 4(T - F_B) = P$	M1 (S+)
	Correct equations but allow any inequality for the = if using μR May evaluate the trig terms Follow through on their (positive) T and condone inclusion of T in an attempt at the upper bound	A1ft
	$T = "10mg" + \frac{1}{2}R_A$ $\left(T = "10mg" + \frac{1}{2}("24mg") = 22mg \right)$ or $T + \frac{4}{5}P + \frac{1}{5}R_B = "45mg"$ or $(T) + \frac{4}{5}P - \frac{1}{5}R_B = "45mg"$	M1
	$"22mg" + \frac{4}{5}P + \frac{1}{5}\left("60mg + \frac{3}{5}P"\right) = "45mg"$	Eliminates R_B from their
		dddM1

Question Number	Scheme		Marks
	$\begin{aligned} &\text{or} \\ &("22mg") + \frac{4}{5}P - \frac{1}{5}\left("60mg + \frac{3}{5}P"\right) = "45mg" \\ &\Rightarrow P = \dots \end{aligned}$	parallel equation/inequality, uses their T and proceeds to find a bounding value for P	
	$P = \frac{1425}{17}mg \text{ or } P = \frac{275}{23}mg$	One correct bounding value for P (either k_1 or k_2 correct and ignore labelling)	A1
	$\begin{aligned} &"22mg" + \frac{4}{5}P + \frac{1}{5}\left("60mg + \frac{3}{5}P"\right) = "45mg" \\ &\text{and} \\ &\frac{4}{5}P - \frac{1}{5}\left("60mg + \frac{3}{5}P"\right) = "45mg" \\ &\Rightarrow P = \dots \text{ and } P = \dots \end{aligned}$	Uses symmetry or restarts on block B with friction acting in the other direction leading to another bounding value for P T must now be 0 for the upper bound, and directions of forces must have been correct.	M1 (S+)
	$\begin{aligned} &\frac{275}{23}mg \leq P \leq \frac{1425}{17}mg \\ &\text{or} \\ &k_1 = \frac{275}{23} \text{ and } k_2 = \frac{1425}{17} \end{aligned}$	Correct range for P or correct values for both k_1 and k_2 including labelling	A1
			(9)
(b)	e.g. $R_{pulley}^2 = T^2 + T^2 - 2T^2 \cos(\alpha + \beta)$ or $R_x = T(\cos \alpha - \cos \beta) = \frac{8}{65}T$ and $R_y = T(\sin \alpha + \sin \beta) = \frac{64}{65}T$ 	Uses the cosine rule to find an expression for the reaction force on the pulley (or the square of this) or other trig methods can be used (e.g use of isosceles triangle).	M1
	Alternatively, resolves tension to find the horizontal and vertical components of R		
	$\begin{aligned} &\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &\left(= \frac{12}{13} \times \frac{4}{5} - \frac{5}{13} \times \frac{3}{5} = \frac{33}{65} \right) \\ &\text{or} \\ &R^2 = \left(\frac{8}{65} \right)^2 T^2 + \left(\frac{64}{65} \right)^2 T^2 \quad \left(= \left(\frac{64}{65} \right)^2 T^2 \right) \end{aligned}$	Uses addition formula to expand $\cos(\alpha + \beta)$ In the alternative, uses Pythagoras with their horizontal and vertical components	dM1
	$R_{pulley}^2 = \frac{64}{65}T^2$	Correct exact expression for the reaction on the	A1

Question Number	Scheme		Marks
	$\Rightarrow R_{pulley} = \frac{8\sqrt{65}}{65}T$ or $\frac{8}{\sqrt{65}}T$	pulley in terms of T	
			(3)
	For (c) and (d) accept less formal language as long as the intention is clear, but there must be an effect and a reason provided for each. In each case award SC B1B0B0 if the effects are all given correctly but no correct justifications.		
(c)(i)	Block B accelerates/moves down the plane as e.g. the total resistance force from P and friction is no longer sufficient to keep it from sliding down the plane.	Effect and explanation given. Must refer to friction in some way.	B1 (S+)
(ii)	Block A accelerates/moves up the plane as e.g. the string is taut, both blocks accelerate at the same rate.	Effect and explanation given. Must reference the string / connected system.	B1 (S+)
(iii)	The reaction force on the pulley increases as T increases.	Effect and explanation given.	B1 (S+)
			(3)
(d)(i)	Block B accelerates/moves up the plane as e.g. P is too large for friction to keep it from sliding up the plane.	Effect and explanation given. Must refer to friction in some way.	B1 (S+)
(ii)	Block A does not move as e.g. maximal friction is larger than the component of block A 's weight down the plane.	Effect and explanation given.	B1 (S+)
(iii)	The reaction force on the pulley is 0 as e.g. T is 0 because the string is slack (and the pulley is modelled as being small).	Effect and explanation given.	B1 (S+)
			(3)
	Award S2 for a solution scoring 15+ marks that is succinct and includes some S+ points (see below). Award S1 for: - a fully correct solution that is succinct but does not mention any S+ points - solution scoring 13+ marks that may be laboured but includes at least 2 S+ points - solution scoring 13+ marks that is succinct but includes at least one S+ point		S2

(Total 18+2 marks)

Notes:

S+ for good reasoning for considering friction acting in opposite directions
S+ for not restarting for friction acting in the other direction
S+ for not including tension for the upper bound
S+ for good reasoning for at least three points in part (c) and/or part (d)
S+ for good reasoning for at least five points in part (c) and part (d)

Question Number	Scheme		Marks
6(a)	$y = \arctan x \Rightarrow x = \tan y$ $\Rightarrow \frac{dx}{dy} = \sec^2 y$ oe	Makes x the subject and differentiates with respect to y , or may differentiate implicitly.	M1
	$\Rightarrow \frac{dx}{dy} = 1 + \tan^2 y = 1 + x^2$ $\Rightarrow \frac{dy}{dx} = \frac{1}{1 + x^2}$ *	cso with no steps omitted. May use other identities.	A1*
			(2)
6(b)(i)	$u = a + b - x \Rightarrow$ $\frac{du}{dx} = -1$ $x = a \rightarrow u = b$ $x = b \rightarrow u = a$ $\int_a^b f(x) dx = - \int_b^a f(a + b - u) du$	Full attempt to replace x with u . Must see a substitution with u used.	M1
	$\int_a^b f(x) dx = - \int_b^a f(a + b - u) du$ $= \int_a^b f(a + b - u) du$ $= \int_a^b f(a + b - x) dx$ *	Convincing argument with fully correct work. Must show the reversal of limits in u and then replace u with x	A1* (S+)
	$\lambda = \frac{1}{2}$	Deduces $\lambda = \frac{1}{2}$. May be embedded.	B1
			(3)
(c)	e.g. $I = \frac{1}{2} \int_{-\pi}^{\pi} \frac{x(1 + \sin x)}{1 + \cos^2 x} + \frac{-x(1 + \sin(-x))}{1 + \cos^2(-x)} dx$	Uses the result from (b) Allow ft on their λ for this mark	M1 (S+)
	e.g. $I = \frac{1}{2} \int_{-\pi}^{\pi} \frac{2x \sin x}{1 + \cos^2 x} dx = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$ *	Convincing argument with fully correct work	A1*
			(2)
Alt (c) using odd property of integrand	$\int_{-\pi}^{\pi} \frac{x}{1 + \cos^2 x} dx = \frac{1}{2} \int_{-\pi}^{\pi} \frac{x}{1 + \cos^2 x} + \frac{-x}{1 + \cos^2(-x)} dx$ $= 0$	Uses the result from (b) to show that $\int_{-\pi}^{\pi} \frac{x}{1 + \cos^2 x} dx = 0$ Allow ft on their λ for this mark	M1
	$I = \int_{-\pi}^{\pi} \frac{x + x \sin x}{1 + \cos^2 x} dx = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$ *	Convincing argument with fully correct work May score SC M1A0 if they argue that $\int_{-\pi}^{\pi} \frac{x}{1 + \cos^2 x} dx = 0$ because the integrand is odd without showing this	A1*

Question Number	Scheme		Marks
			(2)
(d)	$g(-x) = \frac{-x \sin(-x)}{1 + \cos^2(-x)} = \frac{x \sin x}{1 + \cos^2 x} = g(x)^*$	cso with sufficient clarity. No need to see the g's	B1*
			(1)
(e)	$I = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx = 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$	Applies the result from (d) to the result from (c) to rewrite the integral	M1
	$= \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} + \frac{(\pi - x) \sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$	Uses the result from (b)	dM1 (S+)
	$= \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} + \frac{(\pi - x) \sin x}{1 + \cos^2 x} dx$ $= \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} dx$	Uses $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x$ to achieve a common denominator (either of which may be implied) Correct simplified integrand with correct limits	ddM1 (S+) A1
	$u = \cos x \Rightarrow \frac{du}{dx} = -\sin x$ $I = \int_{\{1\}}^{\{-1\}} \frac{-\pi}{1 + u^2} du \left(= \pi \int_{\{-1\}}^{\{1\}} \frac{1}{1 + u^2} du \right)$	Fully substitutes, using a suitable substitution to achieve an integrable form. There is no need to see limits addressed for this mark.	M1 (S+)
	$x = \pi \rightarrow u = -1 \text{ and } x = 0 \rightarrow u = 1$	Correct changing of limits for their suitable substitution on relevant work.	B1
	$I = \pi [\arctan u]_{-1}^1$ $= \pi (\arctan(1) - \arctan(-1))$	Uses the result from (a) to integrate to $\arctan u$ and substitutes changed limits.	dM1
	$= \pi \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) = \frac{\pi^2}{2}$	cso	A1
			(8)
Alt for last 4 marks	Alt: $I = -(\pi) \int_0^{\pi} \frac{-\sin(x)}{1 + \cos^2 x} dx$	Sets up or applies a reverse chain rule using (a). May be implied by a correct integral.	M1 (S+)
	$= -\pi [\arctan(\cos x)]_0^{\pi}$	Correct integral	B1
	$= -\pi (\arctan(\cos(\pi)) - \arctan(\cos(0)))$	Applies the limits	dM1

Question Number	Scheme		Marks
	$= \pi \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) = \frac{\pi^2}{2}$	Cso	A1
(e) Alt 1 for the first 4 marks	$I = \int_{-\pi}^{\pi} \frac{x \sin x}{1 + \cos^2 x} \mathrm{d}x = 2 \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \mathrm{d}x$	Applies the result from (d) to the result from (c) to rewrite the integral	M1
	$I = \int_{-\pi}^0 \frac{2(\pi - u) \sin(\pi - u)}{1 + \cos^2(\pi - u)} (-\mathrm{d}u)$	Uses the substitution $u = \pi - x$	dM1 (S+)
	$I = 2\pi \int_0^{\pi} \frac{\sin u}{1 + \cos^2 u} \mathrm{d}u - 2 \int_0^{\pi} \frac{u \sin u}{1 + \cos^2 u} \mathrm{d}u$ $I = 2\pi \int_0^{\pi} \frac{\sin u}{1 + \cos^2 u} \mathrm{d}u - I$ $I = \int_0^{\pi} \frac{\pi \sin x}{1 + \cos^2 x} \mathrm{d}x$	Uses $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x$ to simplify the integrand and recognises I Correct integrand with correct limits	ddM1 A1
	Award S2 for a solution scoring 13+ marks that is succinct and includes some S+ points (see below). Award S1 for: - a solution scoring 13+ marks that is succinct but does not mention any S+ points - solution scoring 10+ marks that may be laboured but includes at least 2 S+ points - solution scoring 10+ marks that is succinct but includes at least one S+ point		S2
(Total 16+2 marks)			
Notes: S+ for concise work in (b)(i) S+ for concise work using the result from (b)(ii) in either (c) or (e) rather than the result from (b)(i) S+ for concise work in dealing with $\sin(\pi - x) = \sin x$ and $\cos(\pi - x) = -\cos x$ S+ for clear work replacing limits in (e) at the appropriate time / recognising and applying the reverse chain rule.			

Question Number	Scheme			Marks
7(a)	$\ln y = x \ln x$ $\frac{1}{y} \frac{dy}{dx} = \ln x + 1$	$y = e^{x \ln x}$ $\frac{dy}{dx} = e^{x \ln x} \left(x \times \frac{1}{x} + \ln x \right)$	Takes ln of both sides and differentiates with either the LHS or RHS correct or writes as $e^{x \ln x}$ and uses the product and chain rule Fully correct	M1
	$\frac{dy}{dx} = x^x (\ln x + 1)$ or $x^x + x^x \ln x$		Correct derivative in simplest form in terms of x	A1
				(3)
(b)	$y = (x^x)^x \Rightarrow \ln y = x \ln x^x \Rightarrow \frac{1}{y} \frac{dy}{dx} = \ln x^x + x \cdot \frac{x^x (\ln x + 1)}{x^x}$ or $y = x^{(x^2)} \Rightarrow \ln y = x^2 \ln x \Rightarrow \frac{1}{y} \frac{dy}{dx} = 2x \ln x + \frac{x^2}{x}$ or $y = e^{x^2 \ln x} \Rightarrow \frac{dy}{dx} = e^{x^2 \ln x} \left(x^2 \times \frac{1}{x} + 2x \ln x \right)$		Repeats the process, taking ln of both sides and applying the product rule, and their answer to (a) Correct differentiation in any form	M1 (S+)
	e.g. $\frac{dy}{dx} = y [\ln x^x + x (\ln x + 1)]$ $= x (x^x)^x (2 \ln x + 1) = 0$ $\Rightarrow \ln x = -\frac{1}{2} \Rightarrow x = e^{-\frac{1}{2}}$	e.g. for (S+): $2 \ln x + 1 = 0$ $\Rightarrow \ln x = -\frac{1}{2} \Rightarrow x = e^{-\frac{1}{2}}$	Sets = 0 and solves for (non-zero) x May still be in terms of y Correct value for x	dM1 (S+)
	$\left(e^{-\frac{1}{2}}, e^{-\frac{1}{2e}} \right)$ or e.g. $\left(\frac{1}{\sqrt{e}}, \frac{1}{\sqrt[2e]{e}} \right)$		Correct coordinates in a simplified exact form, allow as $x = ..$, $y = ..$ Note 10111 is possible.	A1
				(5)
(c)	$y = x^{(x^x)} \Rightarrow \ln y = x^x \ln x$ $\Rightarrow \frac{1}{y} \frac{dy}{dx} = \frac{x^x}{x} + x^x (\ln x + 1) \ln x$ or e.g. $y = e^{x^x \ln x}$ $\Rightarrow \frac{dy}{dx} = e^{x^x \ln x} \left(x^x \times \frac{1}{x} + x^x (\ln x + 1) \ln x \right)$	Repeats the process, taking ln of both sides and applying the product rule, and their answer to (a) Correct differentiation in any form		M1
	$\Rightarrow \frac{dy}{dx} = x^{(x^x)} \left(x^{x-1} + x^x (\ln x + 1) \ln x \right)$ $= x^{(x^x + x - 1)} (1 + x (\ln x + 1) \ln x)$ $= x^{(x^x + x - 1)} \left(x (\ln x)^2 + x \ln x + 1 \right) *$	Multiplies by y and factorises to the required form. Correct differentiation to the required form.		dM1 (S+) A1*

Question Number	Scheme		Marks
	or $y = e^{x^x \ln x} \Rightarrow \frac{dy}{dx} = x^{(x^x)} (x^{x-1} + x^x (\ln x + 1) \ln x)$ $\Rightarrow \frac{dy}{dx} = x^{(x^x + x - 1)} (x (\ln x)^2 + x \ln x + 1)^*$ i.e. $P(x) = x (\ln x)^2 + x \ln x + 1$ and $Q(x) = x^x + x - 1$	Allow either $(\ln x)^2$ or $\ln^2 x$ May be scored for correct $P(x)$ and $Q(x)$ stated following correct work	
			(4)
(d)	$x \ln x (\ln x + 1) = 0 \Rightarrow x = \dots$	Cancels the 1s, factorises fully, and solves for at least one non-zero value of x or may be implied by $x = 1$ or $x = e^{-1}$	M1
	$x = 1$ and $x = e^{-1}$	Both correct (ignore reference to $x = 0$)	A1
			(2)
(e)	$P'(x) = (\ln x)^2 + \frac{2x \ln x}{x} + \ln x + \frac{x}{x}$ $(= (\ln x)^2 + 3 \ln x + 1)$	Attempts to differentiate $P(x)$ (two terms correct).	M1
	$P'(x) = \left(\ln x + \frac{3}{2} \right)^2 - \frac{5}{4} = 0$ $\Rightarrow \ln x = \frac{-3 \pm \sqrt{5}}{2}$ so $x = e^{\frac{-3 \pm \sqrt{5}}{2}}$	Solves $P'(x) = 0$ for x using correct operations (must reach $x =$)	M1
	$\alpha = e^{\frac{-3 - \sqrt{5}}{2}}$ and $\beta = e^{\frac{-3 + \sqrt{5}}{2}}$	Correct values for α and β (must be correctly labelled)	A1
			(3)
(f)	$P''(x) = \frac{2 \ln x + 3}{x}$ $\Rightarrow P''\left(e^{\frac{-3 + \sqrt{5}}{2}}\right) = \frac{2 \left(\frac{-3 + \sqrt{5}}{2} \right) + 3}{e^{\frac{-3 + \sqrt{5}}{2}}}$ or attempts $P'(1) = 1$ and $P'(e^{-1}) = -1$ with one correct	Differentiates $P'(x)$ (two terms correct) and attempts to find the value of $P''(x)$ at the larger of the two roots of $P'(x) = 0$ Alternatively attempts $P'(1) = 1$ and $P'(e^{-1}) = -1$ with at least one correct. Or shows that $P(\beta) < 1$ as in (g)	M1
	$\Rightarrow P''(\beta) = \sqrt{5} e^{\frac{3 - \sqrt{5}}{2}} > 0 \therefore \text{min at } x = \beta$ or shows $P'(1) = 1 > 0$ and $P'(e^{-1}) = -1 < 0$ and $e^{-1} < e^{\frac{-3 + \sqrt{5}}{2}} < 1 \therefore \text{min at } x = \beta$ or	Correct value for the second derivative at $x = \beta$, shows it is positive and concludes it is a minimum. See alternatives	A1*

Question Number	Scheme		Marks
	shows $P(\beta) < 1$ and states either $P(1) = 1$ or $P(e^{-1}) = 1$ and $P(x)$ is continuous $\therefore \min$ at $x = \beta$		
			(2)
(g)	Since $x > 0$, ($x^k > 0$ for all k and) hence $x^{Q(x)} > 0$ for all $x > 0$	Identifies $x^{Q(x)} > 0$ for $x > 0$	B1 (S+)
	For $\beta = e^{\frac{-3+\sqrt{5}}{2}}$, $P(\beta) = e^{\frac{-3+\sqrt{5}}{2}} (2 - \sqrt{5}) + 1$	Attempts to find the value of $P(\beta)$ (need not be simplified). Allow if done in (f).	M1
	Since $0 < e^{\frac{-3+\sqrt{5}}{2}} < e^0 = 1$ and $-1 < 2 - \sqrt{5} < 0$ we have $-1 < e^{\frac{-3+\sqrt{5}}{2}} (2 - \sqrt{5}) (< 0)$ and hence $0 < P(\beta) (< 1)$ (dM0 if an incorrect β trivialises the problem)	Attempts to show that the value of $P(\beta)$ is greater than 0 by analysing its terms (which may be seen in (f) but must be referenced in (g).	dM1 (S+)
	Since $P(x)$ has a single local minimum point at $x = \beta$, with $P(\beta) > 0$ and $P(x)$ only crosses $y = 1$ once either side of $x = \beta$ and $P(x)$ is a continuous function we conclude the minimum is a global minimum hence $P(x) > 0$ Since also $x^{Q(x)} > 0$, we have $g'(x) > 0$ for $x > 0$ and hence $g(x)$ is an increasing function for all $x > 0$	Complete argument including all relevant points	A1* (S+)
			(4)
	Award S2 for a solution scoring 16+ marks that is succinct and includes some S+ points (see below). Award S1 for: - a fully correct solution that is succinct but does not mention any S+ points - solution scoring 13+ marks that may be laboured but includes at least 2 S+ points - solution scoring 13+ marks that is succinct but includes at least one S+ point		S2

(Total 23+2 marks)

Notes:

S+ for using $y = x^{(x^2)}$ in (b)

S+ for concise work following use of $\frac{dy}{dx} = 0$ to avoid unnecessary factorisation (e.g. not multiplying by y)

S+ for concise work in (c)

S+ for convincing argument that $x^{Q(x)} > 0$ in (g)

S+ for a strong argument as to the value of $P(\beta)$ in (g)

S+ for strong reasoning to complete the argument in (g)